Wasiur Rahman Khuda Bukhsh | Assistant Professor of Mathematics

School of Mathematical Sciences – The University of Nottingham
University Park, Nottingham, NG7 2RD, UK
wasiur.khudabukhsh@nottingham.ac.uk • wasiur.xyz
0000-0003-1803-0470 • wasiur

Professional Experience

School of Mathematical Sciences — **The University of Nottingham, UK**
Assistant Professor — *Oct 2021 – Present*

Mathematical Biosciences Institute & College of Public Health — **The Ohio State University, USA**
President's Postdoctoral Scholar — *Jan 2020 – Sep 2021*

Mathematical Biosciences Institute & College of Public Health — **The Ohio State University, USA**
Postdoctoral Researcher — *Oct 2018 – Dec 2019*

Department of Electrical Engineering and Information Technology — **Technische Universität Darmstadt, Germany**
Research Associate — *Jan 2014 – Sep 2018*

Business Intelligence Unit — **ICICI Bank, Mumbai, India**
Manager, Advanced Analytics — *Jun 2011 – Dec 2013*

Education

University of Nottingham — **UK**
Postgraduate Certificate in Higher Education (PGCHE) — *2023 – 2024*

Advance HE (formerly Higher Education Academy) — **UK**
Fellow (FHEA) — *2023 – Present*

Technische Universität Darmstadt — **Germany**
Ph.D. (Dr. rer. nat.), summa cum laude — *2014 – 2018*
Department of Electrical Engineering and Information Technology

Indian Statistical Institute — **Kolkata, India**
Master of Statistics (M.Stat.) — *2009 – 2011*

University of Calcutta — **Kolkata, India**
B.Sc. (Honours) in Statistics — *2006 – 2009*

Publications

Preprints/Submitted

[1] J. D. Harborne, **Wasiur R. KhudaBukhsh**, and John R. King. Deterministic and stochastic asymptotics for a multiscale epidemic model. 2026. Submitted.

[2] Arnab Ganguly and **Wasiur R. KhudaBukhsh**. Asymptotic analysis of the total quasi-steady state approximation for the Michaelis–Menten enzyme kinetic reactions. 2025. Revision requested.

[3] Yordan P. Raykov, Hengrui Luo, Justin D. Strait, and **Wasiur R. KhudaBukhsh**. Kernel-based estimators for functional causal effects. 2026. Under review.

[4] **Wasiur R. KhudaBukhsh** and Yangrui Xiang. Mixing time for a noisy SIS model on graphs. 2026. Submitted.

[5] Karim S. Elsayed, Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, and Amr Rizk. Stochastic analysis of entanglement-assisted quantum communication channels. 2025. Revision requested.

[6] Olga Izyumtseva and **Wasiur R. KhudaBukhsh**. Local times of self-intersection and sample path properties of Volterra Gaussian processes. 2025. Revision requested.

[7] Arnab Ganguly and **Wasiur R. KhudaBukhsh**. Enzyme kinetic reactions as interacting particle systems: Stochastic averaging and parameter inference. 2026. Revision requested..

Peer-reviewed Journal Publications

[8] Yi Fu, Hye-Won Kang, **Wasiur R. KhudaBukhsh**, Lea Popovic, Grzegorz A. Rempała, and Ruth J. Williams. Fragility in a Togashi-Kaneko stochastic model with mutations. *SIAM Journal on Life Sciences*, 2026. Accepted for publication.

[9] Riccardo Corradin, Luca Danese, **Wasiur R. KhudaBukhsh**, and Andrea Ongaro. Model-based clustering of time-dependent observations with common structural changes. *Statistics and Computing*, 36(7), 2026.

[10] Alexander E. Zarebski, Nefel Tellioglu, Jessica E. Stockdale, Julie A. Spencer, **Wasiur R. KhudaBukhsh**, Joel C. Miller, and Cameron Zachreson. Including frameworks of public health ethics in computational modelling of infectious disease interventions. *Interface Focus*, 15, 2025.

[11] Yushuf Sharker, Zaynab Diallo, **Wasiur R. KhudaBukhsh**, and Eben Kenah. Pairwise accelerated failure time regression models for infectious disease transmission in close contact groups with external sources of infection. *Statistics in Medicine*, 2024.

[12] **Wasiur R. KhudaBukhsh** and Grzegorz A. Rempała. How to *correctly* fit an SIR model to data from an SEIR model? *Mathematical Biosciences*, 2024.

[13] Matthew Wascher, Patrick Schnell, **Wasiur R. KhudaBukhsh**, Mikkel B.M. Quam, Joseph Tien, and Grzegorz Rempała. Estimating disease transmission in a closed population under repeated testing. *Journal of the Royal Statistical Society: Series C (JRSSC)*, 2024.

[14] István Z. Kiss, Luc Berthouze, and **Wasiur R. KhudaBukhsh**. Towards inferring network properties from epidemic data. *Bulletin of Mathematical Biology*, 2024.

[15] **Wasiur R. KhudaBukhsh**, Sat Kartar Khalsa, Eben Kenah, Grzegorz Rempała, and Joseph Tien. COVID-19 dynamics in an Ohio prison. *Frontiers in Public Health*, 2023.

[16] **Wasiur R. KhudaBukhsh**, Caleb Deen Bastian, Matthew Wascher, Colin Klaus, Saumya Yashmohini Sahai, Mark H. Weir, Eben Kenah, Elisabeth Root, Joseph H. Tien, and Grzegorz A. Rempała. Projecting COVID-19 cases and hospital burden in ohio. *Journal of Theoretical Biology*, 561:111404, 2023.

[17] Colin Klaus, Matthew Wascher, **Wasiur R. KhudaBukhsh**, and Grzegorz Rempała. Likelihood-Free Dynamical Survival Analysis applied to the COVID-19 epidemic in Ohio. *Mathematical Biosciences and Engineering*, 20, 2023.

[18] Kai Cui, **Wasiur R. KhudaBukhsh**, and Heinz Koeppl. Hypergraphon mean-field games. *Chaos*, 2022.

[19] **Wasiur R. KhudaBukhsh**, Casper Woroszylo, Grzegorz Rempała, and Heinz Koeppl. A functional central limit theorem for SI processes on configuration model graphs. *Advances in Applied Probability*, 2022.

[20] Colin Klaus, Matthew Wascher, **Wasiur R. KhudaBukhsh**, Joseph H. Tien, Grzegorz A. Rempała, and Eben Kenah. Assortative mixing among vaccination groups and biased estimation of reproduction numbers. *The Lancet Infectious Diseases*, 22:P579–581, 5 2022.

[21] Francesco Di Lauro*, **Wasiur R. KhudaBukhsh***, István Z. Kiss, Eben Kenah, Max Jensen, and Grzegorz Rempała. Dynamic survival analysis for non-markovian epidemic models. *Journal of the Royal Society Interface*, 2022. *Both authors contributed equally and are joint first authors.

[22] Kai Cui, **Wasiur R. KhudaBukhsh**, and Heinz Koeppl. Motif-based mean-field approximation of interacting particles on clustered networks. *Physical Review E*, 105, 4 2022.

[23] Harley Vossler, Pierre Akilimali, Yuhan Pan, **Wasiur R. KhudaBukhsh**, Eben Kenah, and Grzegorz A. Rempała. Analysis of individual-level epidemic data: Study of 2018-2020 ebola outbreak in democratic republic of the congo. *Scientific Reports*, 12, 2022.

[24] Ido Somekh*, **Wasiur R. KhudaBukhsh***, Elisabeth Dowling Root*, Greg Rempala, Eric Simoes, and Eli Somekh. Quantifying the Population-level Effect of COVID-19 Mass Vaccination Campaign in Israel: A Modeling Study. *Open Forum Infectious Diseases*, 2022. *Equal contribution.

[25] **Wasiur R. KhudaBukhsh***, Hye-Won Kang, Eben Kenah, and Grzegorz Rempała. Incorporating age and delay into models for biophysical systems. *Physical Biology*, 18(1), 2021. (*Invited paper).

[26] **Wasiur R. KhudaBukhsh**, Boseung Choi, Eben Kenah, and Grzegorz Rempała. Survival dynamical systems: individual-level survival analysis from population-level epidemic models. *Journal of the Royal Society Interface Focus*, 10(1), 2020.

[27] **Wasiur R. KhudaBukhsh**, Arnab Auddy, Yann Disser, and Heinz Koeppl. Approximate lumpability for Markovian agent-based models using local symmetries. *Journal of Applied Probability*, 56, 9 2019.

[28] Hye-Won Kang*, **Wasiur R. KhudaBukhsh***, Heinz Koeppl, and Grzegorz Rempała. Quasi-steady-state approximations derived from a stochastic enzyme kinetics. *Bulletin of Mathematical Biology*, 81(5):1303–1336, 2019. *joint first authors.

[29] Saumya Yashmohini Sahai, Saket Gurukar, **Wasiur R. KhudaBukhsh**, Srinivasan Parthasarathy, and Grzegorz A. Rempała. A Machine Learning Model for Nowcasting Epidemic Incidence. *Mathematical Biosciences*, 2021.

[30] **Wasiur R. KhudaBukhsh**, Sounak Kar, Bastian Alt, Amr Rizk, and Heinz Koeppl. Generalized cost-based job scheduling in very large cluster systems. *IEEE Transactions on Parallel and Distributed Systems*, 31(11):2594–2604, 2020.

[31] Boseung Choi, Sydney Busch, Dieudonné Kazadi, Benoit Ilunga, Emile Okitolonda, Yi Dai, Robert Lumpkin, Omar Saucedo, **Wasiur R. KhudaBukhsh**, Joseph Tien, Marcel Yotebieng, Eben Kenah, and Grzegorz A. Rempała. Modeling Outbreak Data: Analysis of a 2012 Ebola Virus Disease Epidemic in DRC. *BIOMATH*, 8(2), 2019.

[32] **Wasiur R. KhudaBukhsh**, Amr Rizk, Sounak Kar, and Heinz Koeppl. Provisioning and performance evaluation of parallel systems with output synchronization. *ACM Transactions on Modeling and Performance Evaluation of Computing Systems (TOMPECS)*, 4(1), 3 2019.

[33] Bastian Alt, Markus Weckesser, Christian Becker, Matthias Hollick, Sounak Kar, Anja Klein, Robin Klose, Roland Kluge, Heinz Koeppl, Boris Koldehofe, **Wasiur R. KhudaBukhsh**, Mahdi Mousavi, Martin Pfannemueller, Amr Rizk, Andy Schuerr, and Ralf Steinmetz. Transitions: A protocol-independent view of the future internet. *Proceedings of the IEEE*, 107(4):835–846, 4 2019.

Book Chapters

[34] Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, and Grzegorz A. Rempała. Functional law of large numbers for an epidemic model with random effects. Handbook of Statistics. Elsevier, 2024.

[35] Grzegorz A. Rempała and **Wasiur R. KhudaBukhsh**. Dynamical survival analysis for epidemic modeling. In *Handbook of Visual, Experimental and Computational Mathematics*, pages 1–17. Springer International Publishing, 2023.

Conference Proceedings

[36] Karim Elsayed, **Wasiur R. KhudaBukhsh**, and Amr Rizk. On the Trade-off between Fidelity and Latency for the Quantum Link Layer with few Memories and Entanglement Purification. In *Proceedings of the International Conference on Quantum Communications, Networking, and Computing (QCNC 2024)*, 2024. Best paper award.

[37] Karim Elsayed, **Wasiur R. KhudaBukhsh**, and Amr Rizk. On the Fidelity Distribution of Link-level Entanglements under Purification. In *Proceedings of the IEEE International Conference on Communication (ICC) 2024*, 2024.

[38] Riccardo Corradin, Luca Danese, **Wasiur KhudaBukhsh**, and Andrea Ongaro. Model-based clustering of non-stationary time series with common historical change times. In *Statistical Learning, Sustainability and Impact*, 2023.

[39] **Wasiur R. KhudaBukhsh**, Bastian Alt, Sounak Kar, Amr Rizk, and Heinz Koeppl. Collaborative uploading in heterogeneous networks: Optimal and adaptive strategies. In *IEEE International Conference on Computer Communications (INFOCOM)*, 4 2018. $< 20\%$ acceptance rate. Best-in-Session Presentation Award.

[40] **Wasiur R. KhudaBukhsh**, Amr Rizk, Alexander Frömmgen, and Heinz Koeppl. Optimizing Stochastic Scheduling in Fork-Join Queueing Models: Bounds and Applications. In *IEEE International Conference on Computer Communications (INFOCOM)*, 5 2017. $\sim 20\%$ per cent acceptance rate.

[41] Adrian Šošić, **Wasiur R. KhudaBukhsh**, A. M. Zourbir, and Heinz Koeppl. Inverse reinforcement learning in swarm systems. In *AAMAS Workshop on Transfer in Reinforcement Learning*, May 2017. Available: `http://www.tirl.info/proceedings/2017/SosicEtal-Inverse.pdf`.

[42] Adrian Šošić, **Wasiur R. KhudaBukhsh**, A. M. Zourbir, and Heinz Koeppl. Inverse reinforcement learning in swarm systems. In *International Conference on Autonomous Agents & Multiagent Systems (AAMAS)*, 5 2017. $\sim 26\%$ per cent acceptance rate, Best Paper Award Finalist.

[43] **Wasiur R. KhudaBukhsh**, Julius Rückert, Julian Wulfheide, David Hausheer, and Heinz Koeppl. Analysing and Leveraging Client Heterogeneity in Swarming-based Live Streaming. In *IFIP Networking Conference (IFIP Networking) and Workshops*, 5 2016. $\sim 26\%$ per cent acceptance rate.

[44] Mahdi Mousavi, Hussein Al Shatri, **Wasiur R. KhudaBukhsh**, Heinz Koeppl, and Anja Klein. Cross-Layer QoE-based Incentive Mechanism for Video Streaming in Multi-Hop Wireless Networks. In *IEEE 86th Vehicular Technology Conference (VTC)*, 9 2017.

Thesis and Technical Notes

[45] **Wasiur R. KhudaBukhsh**. *Model reductions for queueing and agent-based systems with applications in communication networks*. PhD thesis, Technische Universität, Darmstadt, 2018. Available at: `http://tuprints.ulb.tu-darmstadt.de/7588/`.

[46] Mark Sinzger-D'Angelo, Heinz Koeppl, and **Wasiur R. KhudaBukhsh**. Bounds on the spectral radius of real-valued non-negative kernels on measurable spaces. 2023. Preprint: `https://arxiv.org/abs/1808.00258`.

Awards & Funding

Awards

- Best Paper Award, International Conference on Quantum Communications, Networking, and Computing (QCNC 2024)
- President's Postdoctoral Scholars Program (PPSP), The Ohio State University (2019)
- Best-in-Session Presentation Award, IEEE INFOCOM, Honolulu, Hawaii (2018)

Research Funding

- **EPSRC EP/Y027795/1** – č79,987.58 (March 2024 – February 2025)

Other Grants & Travel Awards

- Isaac Newton Institute Retreats (2024)
- CM2LA Workshop Grant, South Korea ($7,000)
- LMS Research in Pairs (2024) – č1,200
- MATRIX-Simons Travel Grant – A$2,500
- UoN International Collaboration Fund – č3,150
- HIMR Small Grants – č1,000
- Royal Society International Exchange (2022) – č11,702
- LMS Undergraduate Bursary (2022) – č1,800
- IMS New Researchers Conference (2022)

- AIM SQuaREs Program
- DAAD Travel Grant (2018) – 2,000

Teaching Experience

University of Nottingham, UK

- **MATH3060:** Financial Mathematics (2025–26, with M. Tretyakov)
- **MATH2109:** Probability 3/Markov Chains (Spring 2025, 2026)
- **MATH2010:** Probability Methods and Models (Spring 2022–2024)
- **MATH4045:** Mathematical Finance (2022–2025)

The Ohio State University, USA

- **STAT 4201:** Introduction to Mathematical Statistics (Fall 2019)
- **PUBHBIO7193:** Advanced Stochastic Epidemiology (Summer 2019)

Technische Universität Darmstadt, Germany

- **CMSSB:** Computational Methods for Systems and Synthetic Biology (2016–2017)

Supervision & Mentoring

Summary: 2 postdocs, 3 PhD students, 16 MSc students, 16 BSc students

Postdoctoral Researchers

- Dr. Olga Iziumtseva (British Academy Fellowship), 2023–2025
- Dr. Yangrui Xiang, 2024–2025

PhD Students

- Kushankur Dutta (2023–2027 expected)
- Wesam El Buaishi (2023–2026 expected)
- James Harborne (2022–2026 expected)

Presentations

Summary: 56 invited talks, 28 contributed talks, 7 poster presentations

Selected Invited Talks (2023–2026)

- University of Strathclyde (2026)
- IIT Mandi, India (2026)
- RSS Annual Meeting (2025)
- SPA 2025, Wrocaw, Poland
- BIRS-CMO Workshop (2025)
- Louisiana State University (2024)
- SMB Annual Meeting, Seoul (2024)
- University of Leeds (2023)
- ICIAM, Tokyo (2023)
- Georgia Tech (2023)

Service & Outreach

Professional Committees

- Applied Probability Section Committee, Royal Statistical Society

Editorial Service

- Guest Associate Editor, JRSS Series B (2023)
- Guest Editor, Mathematical Biosciences and Engineering

PhD Thesis Examiner

- Internal: L. Critcher (Nottingham, 2025)
- External: H.C. Yip (Politecnico di Torino, 2024), D. Keliger (Budapest, 2025)

Grant Review

- EPSRC (UKRI), ZonMw (Netherlands) – 2024

Conference Organization (Selected)

- BIRS Workshop on Pandemic Epidemiology (2025)
- CRN Satellite Workshop, South Korea (2024)
- RSS Epidemic Modelling Workshop (2023)
- BIRS Workshop "Preparing for the Next Pandemic" (2022)

Referee Service

Journals: Electronic Journal of Probability, Stochastic Processes & their Applications, SIAM J. Appl. Dyn. Sys., Biometrika, J. Math. Biology, Bull. Math. Biology, IEEE Trans. Automatic Control, IEEE Trans. Networking, and others.

Technical Skills

Programming
- C, Python, R, Julia, SQL

Software
- MATLAB, SAS, LaTeX

Languages
- English, Bengali, Hindi, German (B1)

Media Highlights

- **SIAM DSWeb** – Student Feature (Jan 2021)
- **OSC News** – COVID-19 Projection Models Equip Ohioans
- **IDI News** – Far from Trivial: Path to OSU/IDI COVID-19 Modeling Team
- **STEAM Spotlight** – Dr. Wasiur Rahman KhudaBukhsh

References

Available upon request.

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